# Phenomenological Consequences and Experimental Predictions of an Emergent Vacuum Pseudoscalar Mode

Erick Sangalang
ORCID: 0009-0009-4039-2815

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**Abstract**

We present a phenomenological investigation of experimentally testable consequences arising from a modified electrodynamic framework in which the vacuum possesses emergent mesoscopic structure. The model introduces an effective pseudoscalar order parameter coupled to electromagnetism through a gauge-invariant interaction term. The use of a pseudoscalar degree of freedom is required by the parity properties of the interaction $\phi F_{\mu\nu}\tilde{F}^{\mu\nu} \propto \phi \mathbf{E} \cdot \mathbf{B}$. We examine several observable signatures of the framework, including temperature-dependent vacuum birefringence, delayed electromagnetic emission in resonant systems, modified electromagnetic propagation, and scaling relations relevant to high-$Q$ cavity experiments. A phenomenological damping term is introduced to model finite relaxation times and delayed re-radiation. Approximate experimental constraints, linear-response behavior, and parameter-space estimates are discussed. The framework is formulated as an effective phenomenological theory intended to explore possible emergent vacuum dynamics at laboratory-accessible scales, rather than as a completed microscopic theory.

## 1 Introduction

Classical electrodynamics describes the vacuum as a passive background characterized by constant electromagnetic permittivity and permeability. In this description, the vacuum transmits electromagnetic fields but does not itself possess dynamically accessible macroscopic structure. Modern theoretical physics has nevertheless demonstrated that the vacuum is not entirely trivial. Quantum electrodynamics predicts vacuum polarization effects arising from virtual particle fluctuations, while several theoretical frameworks, including axion electrodynamics, emergent gravity proposals, and condensed-matter-inspired spacetime analogies, suggest that the vacuum may exhibit richer internal structure under certain conditions.

The purpose of the present work is not to propose a complete microscopic theory of vacuum structure. Instead, we investigate the phenomenological consequences that would arise if the vacuum possessed emergent mesoscopic degrees of freedom capable of coupling dynamically to electromagnetic fields.

The framework considered here introduces an effective pseudoscalar order parameter field coupled to electromagnetism through a gauge-invariant interaction term. The resulting phenomenology predicts several experimentally testable effects, including:

1. temperature-dependent vacuum birefringence,

2. delayed electromagnetic emission following resonant excitation,

3. modified electromagnetic propagation in strong-field environments,
4. resonant enhancement effects in high-$Q$ cavity systems.

The framework is intentionally formulated conservatively as an effective phenomenological model rather than a claim of a completed microscopic theory.

# 2 Scope and Objectives

The objectives of the present framework are:

1. to preserve gauge invariance and charge conservation,
2. to introduce an effective emergent vacuum degree of freedom,
3. to derive experimentally testable electromagnetic consequences,
4. to identify measurable laboratory signatures capable of falsifying the model.

The present work does not attempt:

- to derive a complete microscopic theory of the vacuum,
- to replace quantum electrodynamics,
- to propose free-energy extraction,
- or to claim a solution to cosmological dark energy.

# 3 Effective Field-Theory Framework

## 3.1 Lagrangian Formulation

We consider the effective Lagrangian density

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi) + g\phi F_{\mu\nu}\tilde{F}^{\mu\nu}. \tag{1}$$

Here $F_{\mu\nu}$ is the electromagnetic field tensor, $\tilde{F}^{\mu\nu}$ is the dual electromagnetic tensor, $\phi(x,t)$ is an effective pseudoscalar order parameter, and $g$ is a phenomenological coupling constant. In natural units, where $\hbar = c = 1$, the coupling has mass dimension

$$[g] = \text{energy}^{-1}, \tag{2}$$

assuming $\phi$ carries canonical mass dimension one. This dimensional statement is useful when comparing the model with axion-like-particle searches and other effective electrodynamic constraints.

The theory preserves gauge invariance because the interaction depends only on gauge-invariant field combinations. The pseudoscalar character of $\phi$ is required because

$$F_{\mu\nu}\tilde{F}^{\mu\nu} \propto \mathbf{E}\cdot\mathbf{B}, \tag{3}$$

which is odd under parity.

### 3.2 Field Equations

Applying the Euler-Lagrange equations yields

$$\partial_\mu \partial^\mu \phi + \frac{dV}{d\phi} = g F_{\mu\nu} \tilde{F}^{\mu\nu}. \tag{4}$$

Using the identity $F_{\mu\nu} \tilde{F}^{\mu\nu} \propto \mathbf{E} \cdot \mathbf{B}$, this becomes

$$\Box \phi + \frac{dV}{d\phi} = g \mathbf{E} \cdot \mathbf{B}. \tag{5}$$

To describe finite relaxation times and delayed emission, we introduce a phenomenological damping term:

$$\Box \phi + \Gamma \partial_t \phi + \frac{dV}{d\phi} = g \mathbf{E} \cdot \mathbf{B}. \tag{6}$$

Here $\Gamma$ is an effective damping rate. The corresponding relaxation time is

$$\tau_\phi \sim \Gamma^{-1}. \tag{7}$$

This term is not intended as a microscopic derivation of dissipation. Rather, it parameterizes the finite lifetime of the emergent mode in the same way that damping parameters are commonly introduced in effective linear-response models.

### 3.3 Modified Maxwell Equations

The modified Maxwell equations take the approximate form

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} + g \mathbf{B} \cdot \nabla \phi, \tag{8}$$

and

$$\nabla \times \mathbf{B} - \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} = \mu_0 \mathbf{J} + g \left( \mathbf{E} \times \nabla \phi + \mathbf{B} \partial_t \phi \right). \tag{9}$$

Charge conservation remains intact:

$$\partial_\mu J^\mu = 0. \tag{10}$$

### 3.4 Lorentz-Symmetry Considerations

The covariant form of the Lagrangian is Lorentz invariant at the effective-field-theory level. However, specific experimental configurations, such as static magnetic fields, driven cavities, or cryogenic boundary conditions, select preferred laboratory frames. The present model therefore does not require fundamental Lorentz violation. Any apparent symmetry breaking in the proposed experiments is interpreted as configuration-induced rather than a property of the underlying effective interaction.

A viable parameter regime must simultaneously permit observable resonant collective effects while remaining consistent with existing null results in precision electrodynamics. This is a central constraint on the framework.

# 4 Thermodynamic Interpretation

## 4.1 Statistical Interpretation of the Vacuum

The pseudoscalar field $\phi(x,t)$ is interpreted as a coarse-grained order parameter associated with mesoscopic vacuum configurations. The vacuum is modeled phenomenologically as a statistical ensemble characterized by an effective partition function

$$Z = \int d\phi \exp[-\beta H(\phi)], \tag{11}$$

where

$$\beta = \frac{1}{k_B T}. \tag{12}$$

## 4.2 Effective Potential and Entropic Mass Scale

Expanding around equilibrium,

$$V(\phi) \approx \frac{1}{2} m_\phi^2(T)\phi^2 + \lambda\phi^4 + \cdots . \tag{13}$$

Assuming the free energy takes the form

$$F(\phi, T) = U(\phi) - TS(\phi), \tag{14}$$

and expanding the entropy around equilibrium,

$$S(\phi) \approx S_0 - \alpha\phi^2 + \cdots , \tag{15}$$

we obtain

$$F(\phi, T) \approx U(\phi) + \alpha T\phi^2. \tag{16}$$

This implies the phenomenological scaling relation

$$m_\phi^2(T) \propto T. \tag{17}$$

This temperature dependence is a key experimental discriminator of the framework.

# 5 Linear Response and Resonant Dynamics

In the linear regime, the pseudoscalar equation of motion may be written as

$$\left(\Box + \Gamma\partial_t + m_\phi^2\right)\phi = g\mathbf{E}\cdot\mathbf{B}. \tag{18}$$

For a harmonic electromagnetic drive with angular frequency $\omega$, the response can be represented by an effective susceptibility

$$\chi_\phi(\omega) = \frac{1}{m_\phi^2 - \omega^2 - i\Gamma\omega}. \tag{19}$$

The induced pseudoscalar excitation is therefore approximately

$$\phi(\omega) \sim g\chi_\phi(\omega)(\mathbf{E}\cdot\mathbf{B}). \tag{20}$$

This form predicts resonant enhancement when

$$\omega \approx m_\phi, \tag{21}$$

with a linewidth controlled by $\Gamma$. The phase lag implied by the imaginary component of $\chi_\phi$ provides a natural mechanism for delayed electromagnetic re-radiation after the driving field is removed.

The characteristic delayed-emission timescale is expected to scale as

$$\tau_\phi \sim \Gamma^{-1}. \tag{22}$$

Depending on the effective damping rate, possible laboratory signatures could appear over nanosecond, microsecond, or millisecond timescales. A detector-level treatment would be required to identify realistic integration windows and background-rejection protocols.

# 6 Phenomenological Consequences

## 6.1 Temperature-Dependent Vacuum Response

In the static linear regime,

$$\phi \sim \frac{g\mathbf{E}\cdot\mathbf{B}}{m_\phi^2(T)}. \tag{23}$$

Using the temperature scaling relation $m_\phi^2(T) \propto T$, we obtain

$$\phi \propto \frac{g\mathbf{E}\cdot\mathbf{B}}{T}. \tag{24}$$

## 6.2 Vacuum Birefringence

One experimentally accessible consequence is a temperature-dependent refractive-index shift:

$$\Delta n \propto \frac{g^2 B_0^2}{T}. \tag{25}$$

This differs from standard QED vacuum birefringence by the explicit effective-temperature dependence. The strongest phenomenological distinction would therefore arise from controlled measurements of birefringence as a function of magnetic field strength and cryogenic temperature.

## 6.3 Delayed Electromagnetic Emission

The model predicts the possibility of temporary energy transfer between electromagnetic fields and the pseudoscalar vacuum mode. During resonant excitation, a portion of electromagnetic energy may transiently populate the pseudoscalar sector. After removal of the external driving field, the stored excitation may partially re-radiate into electromagnetic modes.

This leads to the prediction of delayed electromagnetic emission. The effect should depend on the damping rate $\Gamma$, cavity quality factor $Q$, electromagnetic field amplitudes, and the degree of resonance matching between the driven electromagnetic mode and the effective pseudoscalar mode.

### 6.4 Resonant Scaling Relations

Approximate scaling behavior for resonant cavity systems may be estimated phenomenologically as

$$P_{\text{out}} \sim g^2 B_0^2 E^2 V \omega Q. \tag{26}$$

For representative laboratory parameters,

$$B_0 \sim 1 - 10 \text{ T}, \tag{27}$$
$$E \sim 10^5 \text{ V/m}, \tag{28}$$
$$V \sim 10^{-4} \text{ m}^3, \tag{29}$$
$$\omega \sim 10^9 \text{ s}^{-1}, \tag{30}$$
$$Q \sim 10^4, \tag{31}$$

the estimated signal range is approximately

$$P_{\text{out}} \sim 10^{-18} - 10^{-12} \text{ W}, \tag{32}$$

subject to the assumed value of $g$ and the normalization convention used for $\phi$.

The leading falsifiable scaling predictions are:

$$P_{\text{out}} \propto B_0^2, \tag{33}$$
$$P_{\text{out}} \propto E^2, \tag{34}$$
$$P_{\text{out}} \propto V, \tag{35}$$
$$P_{\text{out}} \propto \omega, \tag{36}$$
$$P_{\text{out}} \propto Q. \tag{37}$$

If the thermal scaling of the effective mass is correct, then the static response also predicts

$$\Delta n \propto T^{-1}. \tag{38}$$

Failure of these scaling relations under controlled experimental conditions would falsify the phenomenological framework.

## 7 Experimental Configurations

### 7.1 Parameter-Space Visualization

The phenomenological parameter space may be represented in terms of the coupling strength $g$ and effective temperature $T$. The model predicts enhanced sensitivity at lower temperature, stronger magnetic field, stronger electromagnetic driving, and larger cavity quality factor.

A representative parameter-space diagram should use a log-log plot with effective temperature $T$ on the horizontal axis and coupling strength $g$ on the vertical axis. Suggested regions include:

- excluded or constrained regions from precision electrodynamics,
- projected sensitivity from high-$Q$ cavity experiments,
- projected sensitivity from cryogenic polarimetry,

- experimentally accessible regions satisfying $P_{\text{out}} > P_{\text{min}}$.

The detection boundary may be estimated from

$$P_{\text{min}} \sim 10^{-18}\ \text{W} \tag{39}$$

and

$$P_{\text{out}} \sim g^2 B_0^2 E^2 V \omega Q. \tag{40}$$

Representative log-log parameter-space plot.
Horizontal axis: effective temperature $T$.
Vertical axis: coupling strength $g$.
Shaded regions: excluded, constrained, and experimentally accessible.
Detection boundary estimated from $P_{\text{out}} = P_{\text{min}}$.

Figure 1: Schematic parameter space for the phenomenological model. Lower temperatures and stronger effective couplings increase the predicted signal. A completed version should include quantitative exclusion regions from precision electrodynamics and projected sensitivity curves for resonant-cavity and polarimetry experiments.

### 7.2 Predicted Scaling Behavior

| Observable | Scaling Relation | Experimental Handle |
|---|---|---|
| Pseudoscalar response | $\phi \propto g\mathbf{E} \cdot \mathbf{B}/T$ | Field alignment, temperature |
| Vacuum birefringence | $\Delta n \propto g^2 B_0^2/T$ | Magnetic field, cryogenic operation |
| Delayed emission power | $P_{\text{out}} \sim g^2 B_0^2 E^2 V \omega Q$ | Cavity volume, field strength, $Q$ |
| Relaxation time | $\tau_\phi \sim \Gamma^{-1}$ | Time-domain detection |
| Detectability threshold | $P_{\text{out}} > P_{\text{min}}$ | Detector noise floor |

Table 1: Leading phenomenological scaling relations for experimentally testable signatures of emergent vacuum structure.

### 7.3 Resonant Cavity Configuration

A conceptually simple experimental configuration consists of:

1. a high-$Q$ microwave cavity,
2. a strong static magnetic field,
3. a driven resonant electromagnetic mode,
4. delayed-emission detection instrumentation.

The experimental sequence is:

High-$Q$ resonant cavity with static field $B_0$.
RF drive excites a resonant electromagnetic mode.
A delayed detector searches for re-radiated emission after the drive is removed.

Figure 2: Conceptual resonant-cavity configuration. A driven electromagnetic mode interacts with an effective pseudoscalar vacuum excitation in the presence of a static magnetic field $B_0$. After the driving field is removed, delayed electromagnetic emission is searched for at the detector.

1. drive the cavity near resonance,
2. excite the coupled pseudoscalar mode,
3. remove the driving field,
4. search for delayed electromagnetic emission over a time window determined by $\tau_\phi$.

### 7.4 Precision Polarimetry

A second experimental pathway involves precision laser polarimetry. The proposed signature is magnetic-field-dependent birefringence with explicit temperature scaling. Cryogenic operation would maximize sensitivity by enhancing coherent pseudoscalar response if $m_\phi^2(T) \propto T$.

A key experimental discriminator is whether any observed birefringence follows the predicted scaling

$$\Delta n \propto B_0^2 T^{-1}. \tag{41}$$

Standard QED vacuum birefringence does not predict this same effective-temperature dependence, making temperature variation an important test of the present framework.

## 8 Experimental Constraints and Falsifiability

The model could be constrained or ruled out if:

1. no temperature dependence is observed in precision birefringence experiments,
2. delayed electromagnetic emission is absent at predicted sensitivity levels,
3. cavity scaling relations fail experimentally,
4. precision electrodynamic measurements exclude the required coupling parameter space,
5. the observed signal fails to scale as $B_0^2$, $E^2$, $Q$, or $T^{-1}$ where applicable.

The framework therefore generates experimentally testable predictions rather than purely interpretive speculation.

Existing experimental programs provide important reference points. Precision polarimetry experiments such as PVLAS constrain magnetic-field-induced optical effects in vacuum. Axion and axion-like-particle searches, including ADMX and CAST, constrain pseudoscalar couplings to electromagnetism over portions of parameter space. High-$Q$ superconducting cavity experiments also provide relevant sensitivity benchmarks for weak electromagnetic signals.

Although the present model is not identical to a conventional axion-like-particle framework, any viable realization must remain compatible with these null results. Future versions of the model should therefore include a quantitative comparison between the phenomenological coupling $g$ and existing bounds on pseudoscalar-photon interactions.

# 9 Relation to Existing Theories

## 9.1 Axion Electrodynamics

The coupling term

$$\phi F_{\mu\nu}\tilde{F}^{\mu\nu} \tag{42}$$

resembles interactions appearing in axion electrodynamics. However, conventional axion theories interpret the pseudoscalar field as a fundamental particle or axion-like particle. In contrast, the present framework treats the pseudoscalar mode phenomenologically as an emergent mesoscopic vacuum degree of freedom.

Unlike conventional axion frameworks, the present model does not require a fundamental particle excitation with a fixed microscopic mass. The effective mode instead emerges phenomenologically from collective vacuum organization at mesoscopic scales. Its effective mass, damping rate, and response function are treated as experimentally constrained phenomenological quantities.

## 9.2 Quantum Electrodynamics

In standard QED, nonlinear vacuum corrections arise from virtual particle fluctuations and are effectively local at low energies. The present framework instead introduces an effective dynamical pseudoscalar mode capable of mediating energy exchange over finite timescales.

The model must therefore be interpreted as an additional phenomenological sector, not as a replacement for QED. Any allowed parameter regime must reproduce existing precision electrodynamic results in the limit of weak coupling, large damping, or inaccessible resonance conditions.

## 9.3 Emergent and Condensed-Matter Analogies

The framework also shares conceptual similarities with emergent descriptions in condensed matter systems, where collective order parameters can produce long-wavelength effective fields that are not fundamental microscopic degrees of freedom. The present work does not specify the microscopic origin of these configurations, which remains an open theoretical question.

# 10 Cosmological Implications (Speculative)

The pseudoscalar field contributes to the stress-energy tensor according to

$$T_{\mu\nu} = \partial_\mu\phi\partial_\nu\phi - g_{\mu\nu}\left[\frac{1}{2}(\partial\phi)^2 - V(\phi)\right]. \tag{43}$$

In the regime

$$V(\phi) \gg (\partial_t\phi)^2, \tag{44}$$

the equation of state approaches

$$p \approx -\rho. \tag{45}$$

These implications remain speculative and require full cosmological modeling. No claim is made that the present phenomenological pseudoscalar mode constitutes the observed cosmological dark energy. At most, the model suggests that emergent vacuum degrees of freedom could motivate future studies of effective stress-energy contributions in larger-scale settings.

## 11 Limitations and Open Questions

Several important limitations remain:

1. the pseudoscalar mode lacks a microscopic derivation,
2. the coupling constant $g$ must be experimentally constrained,
3. compatibility with precision electrodynamic experiments requires detailed comparison with existing bounds,
4. detector-level modeling has not yet been performed,
5. the damping rate $\Gamma$ remains phenomenological,
6. the relationship between effective temperature and physical laboratory temperature requires further clarification,
7. cosmological implications remain speculative.

The framework should therefore be interpreted as an exploratory phenomenological model rather than a completed microscopic theory.

## 12 Conclusion

We have examined the phenomenological consequences of a modified electrodynamic framework in which the vacuum possesses emergent mesoscopic structure represented by a pseudoscalar order parameter. The framework preserves gauge invariance while introducing experimentally testable electromagnetic signatures, including:

- temperature-dependent vacuum birefringence,
- delayed electromagnetic emission,
- modified propagation effects,
- resonant cavity scaling behavior,
- and finite relaxation dynamics governed by an effective damping rate.

These effects provide experimentally accessible pathways for validation or falsification. The addition of a linear-response description allows the model to make sharper predictions regarding resonance, linewidth, phase lag, delayed emission, and detector timescales.

Future work will focus on microscopic statistical models, detector-level cavity simulations, precision experimental constraints, quantitative parameter-space exclusion plots, and possible extensions to larger-scale physical systems.

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